

2.4.6

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Question

If $\mathbf{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\mathbf{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, show that the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ are perpendicular.

Solution - Part 1

Two vectors are perpendicular if their dot product is zero. We need to show that $(\mathbf{a} + \mathbf{b})^\top (\mathbf{a} - \mathbf{b}) = 0$.

First, we find the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$. The given vectors can be written in column form as:

$$\mathbf{a} = \begin{pmatrix} 5 \\ -1 \\ -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ -5 \end{pmatrix} \quad (1)$$

Solution - Part 2

Now, we calculate the sum and difference:

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 5 + 1 \\ -1 + 3 \\ -3 - 5 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ -8 \end{pmatrix} \quad (2)$$

$$\mathbf{a} - \mathbf{b} = \begin{pmatrix} 5 - 1 \\ -1 - 3 \\ -3 - (-5) \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} \quad (3)$$

Next, we compute the dot product of these two resultant vectors:

$$\begin{aligned} (\mathbf{a} + \mathbf{b})^\top (\mathbf{a} - \mathbf{b}) &= \begin{pmatrix} 6 & 2 & -8 \end{pmatrix} \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} \\ &= (6)(4) + (2)(-4) + (-8)(2) \\ &= 24 - 8 - 16 \\ &= 0 \end{aligned} \quad (4)$$

Conclusion

Since the dot product is zero, the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular. Hence, proved.

C Code - Calculation Function

```
// File: vector_ops.c
float check_perpendicularity(
    float ax, float ay, float az,
    float bx, float by, float bz) {

    // Calculate p = a + b
    float px = ax + bx;
    float py = ay + by;
    float pz = az + bz;

    // Calculate q = a - b
    float qx = ax - bx;
    float qy = ay - by;
    float qz = az - bz;

    // Return the dot product of p and q
    return (px * qx) + (py * qy) + (pz * qz);
}
```

Python Code - Setup and C Library Import

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes
import os

# Load the compiled C library
lib_name = 'vector_ops.so'
c_lib = ctypes.CDLL(os.path.abspath(lib_name))

# Define C function argument and return types
c_lib.check_perpendicularity.argtypes = [ctypes.c_float] * 6
c_lib.check_perpendicularity.restype = ctypes.c_float
```

Python Code - Vector Definition and Calculation

```
# Define vectors and call C function
a = np.array([5, -1, -3], dtype=np.float32)
b = np.array([1, 3, -5], dtype=np.float32)
p = a + b
q = a - b

# Call the C function to get the dot product
dot_product = c_lib.check_perpendicularity(*a, *b)
print(f"Dot product from C: {dot_product}")
if abs(dot_product) < 1e-6:
    print("Vectors are perpendicular.")
```


Python Code - Plotting Logic

```
# Generate the 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
# Dictionary of vectors to plot
vectors = {
    '$\\vec{a}$': (a, 'green'),
    '$\\vec{b}$': (b, 'orange'),
    '$\\vec{a}+\\vec{b}$': (p, 'blue'),
    '$\\vec{a}-\\vec{b}$': (q, 'red')
}
origin = [0, 0, 0]

for name, (vec, color) in vectors.items():
    ax.quiver(*origin, *vec, color=color,
              arrow_length_ratio=0.1, label=name)
    ax.scatter(*vec, color=color, s=50)
    ax.text(vec[0], vec[1], vec[2],
            f'({vec[0]},{vec[1]},{vec[2]})')
```

Python Code - Final Plotting Details

```
# Plotting settings
ax.set_xlabel('X Axis')
ax.set_ylabel('Y Axis')
ax.set_zlabel('Z Axis')
ax.set_title('3D Plot of Vectors and Endpoints')
ax.legend()
ax.grid(True)
ax.view_init(elev=20, azimuth=25) # Set view angle
plt.show()
```

Generated 3D Plot

